Lab Program number.28.. Date:...................

Title: Basic Linux Command

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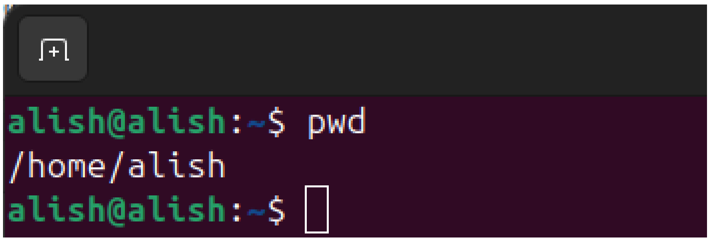
1. pwd

(Print working directory) displays the full path of the current directory we are in.

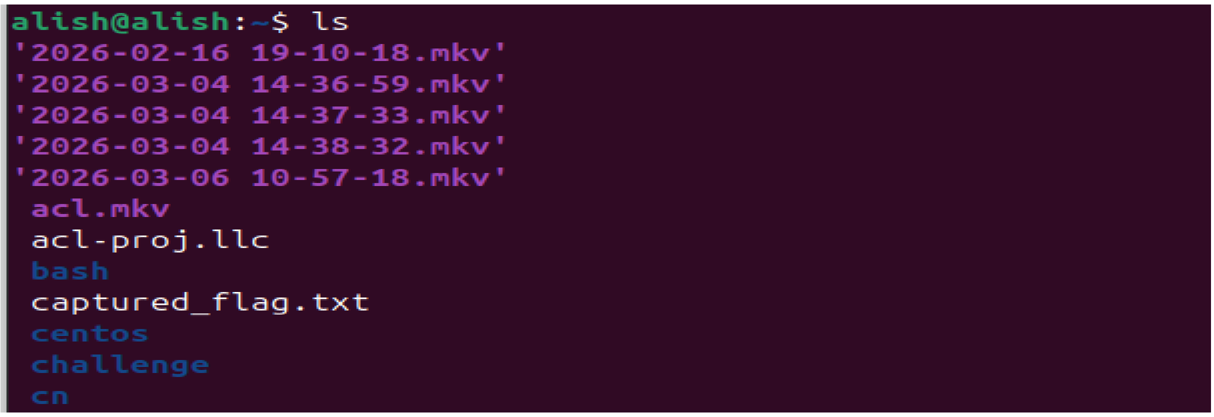
Command : pwd



2. **ls**

List all files and directory in the current location.

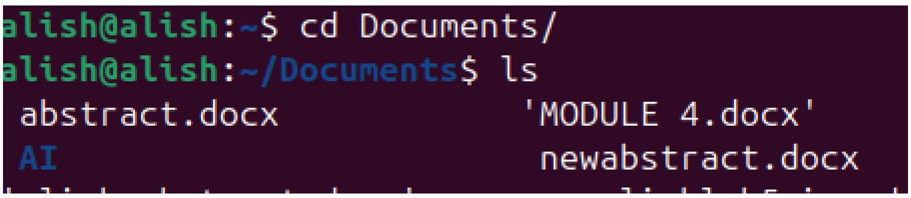
Command: ls



3. **cd (change directory)**

Used to navigate between the folders (cd Document,cd Download)

Command : cd Document



4. **mkdir (make directory)**

Creates a new folder or a directory

Command: mkdir new

```
alish@alish:~$ mkdir new
alish@alish:~$ ls new/
alish@alish:~$
```

5. **rmdir(remove directory)**

Used to delete a empty folder or directory

Command : rmdir new

```
alish@alish:~$ rmdir new/
alish@alish:~$ ls new
ls: cannot access 'new': No such file or dire
alish@alish:~$
```

6. **touch**

Create a new empty file also used to update the timestamp of a existing of file.

Command: touch hello.txt

```
alish@alish:~$ mkdir new
alish@alish:~$ cd new/
alish@alish:~/new$ touch hello.txt
alish@alish:~/new$ ls
hello.txt
alish@alish:~/new$
```

7. **cp(copy)**

Copy files or folder from one location to another

Command : cp hello.txt into.txt

```
alish@alish:~/new$ cp hello.txt into.txt
alish@alish:~/new$ ls
hello.txt  into.txt
alish@alish:~/new$
```

8. **mv(move)**

Move file or folder from one location to another

Command: mv ~/source ~/destination

```
alish@alish:~$ mkdir new1
alish@alish:~$ mv ~/new/hello.txt ~/new1/
alish@alish:~$ cd new1
alish@alish:~/new1$ ls
hello.txt
alish@alish:~/new1$
```

9. **Rm(remove)**

Delete the files and its content

Command – rm filename

– rm –r filename or folder_name

```
alish@alish:~/new1$ ls
hello.txt
alish@alish:~/new1$ rm hello.txt
alish@alish:~/new1$ ls
alish@alish:~/new1$
```

10. **cat(concatenate)**

Displays the entire content of a file on the terminal screen

Command : – cat hello.txt

```
alish@alish:~/new$ cat into.txt
apple
ball
fun
fifa
nepal
happy
messi
banana
spiderman
alish@alish:~/new$
```

11. **head**

head prints the first line desired by the user (eg head –n)

Command: head –n 4 filename

```
alish@alish:~/new$ head -n 4 into.txt
apple
ball
fun
fifa
alish@alish:~/new$
```

12. **tail**

tail prints the last line of the file (eg. tail –n)

Command: tail –n 3 filename

```
alish@alish:~/new$ tail -n 3 into.txt
messi
banana
spiderman
alish@alish:~/new$
```

13.**less**

Unlike cat shows content of a file , less opens the file in an interactive file and you can scroll the content from mouse or arrow key of keyboard

Command : less filename

```
apple
ball
fun
fifa
nepal
happy
messi
banana
spiderman
into.txt (END)
```

```
alish@alish: ~/new
apple
ball
fun
fifa
nepal
happy
messi
banana
spiderman
```

14. **grep (Global Regular Expression Print)**

Grep is a powerful tool used to search text or pattern from file often used with pip | to filter out the output commands

Command : cat into.txt | grep “fifa”

```
alish@alish:~/new$ cat into.txt | grep "fifa"
fifa
alish@alish:~/new$ cat into.txt | grep -n "fifa"
4:fifa
alish@alish:~/new$
```

15. **chmod**

This command modifies the access permissions of a file or directory. In Linux, permissions are divided into **Read (r)**, **Write (w)**, and **Execute (x)** for the **User**, **Group**, and **Others**.

Command : chmod 754 into.txt

```
alish@alish:~/new$ chmod 754 into.txt
alish@alish:~/new$ ls -l into.txt
-rwxr-xr-- 1 alish alish 57 Mar 19 21:45 into.txt
alish@alish:~/new$
```

16. **chown (Change Owner)**

chown is used to change the owner or the group ownership of a file. This is a critical command for system administration when moving files between different user accounts.

Command : sudo chown alish:alish into.txt

```
alish@alish:~/new$ sudo chown alish:alish into.txt
alish@alish:~/new$ ls -l into.txt
-rwxr-xr-- 1 alish alish 57 Mar 19 21:45 into.txt
alish@alish:~/new$
```

17. **sudo (SuperUser Do)**

sudo allows a permitted user to execute a command as the superuser (root) or another user. It is the gatekeeper for tasks that require administrative privileges, like installing software or editing system files.

Command: cd /etc/sudores.d

```
alish@alish:~$ cd /etc/sudoers.d/
alish@alish:/etc/sudoers.d$ ls
README
alish@alish:/etc/sudoers.d$ cat README
cat: README: Permission denied
```

18. **top**

The top command provides a dynamic, real-time view of a running system. It displays system summary information and a list of processes currently being managed by the Linux kernel.

Command: top

```
alish@alish:~$ top

top - 22:19:22 up  3:38,  1 user,  load average: 1.40, 1.42, 1.29
Tasks: 275 total,   1 running, 272 sleeping,   2 stopped,   0 zombie
%Cpu(s):  9.6 us,  2.9 sy,  0.0 ni, 21.5 id, 66.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3696.1 total,    214.2 free,   2468.0 used,   1403.0 buff/cache
MiB Swap:   3695.0 total,   1817.4 free,   1877.6 used.   1228.1 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   6437 alish     20   0   56.7g 180824  53024 S  12.2   4.8  28:25.36 editors+
  19629 alish     20   0   56.7g 233480  66044 S  11.9   6.2  14:17.18 editors+
   4982 alish     20   0 4675476 129680  56624 S   8.9   3.4  10:00.68 gnome-s+
  14937 alish     20   0  709872  51972  36056 S   7.3   1.4   0:39.78 gnome-t+
```

19. **df (Disk Free)**

This command displays the amount of disk space available on the file systems. It helps you monitor how much storage is being used by your hard drives or partitions.

Command: df & df –h

```
alish@alish:~$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
tmpfs             378476     2564    375912   1% /run
/dev/sda2      478512880 89738444 364393868  20% /
tmpfs            1892376    20380   1871996   2% /dev/shm
tmpfs               5120        8      5112   1% /run/lock
efivarfs              24       13         7  65% /sys/firmware/efi/efivars
/dev/sda1        1098628     6288   1092340   1% /boot/efi
tmpfs             378472      152    378320   1% /run/user/1000
alish@alish:~$ 
```

```
alish@alish:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           370M  2.6M  368M   1% /run
/dev/sda2       457G   86G  348G  20% /
tmpfs           1.9G   20M  1.8G   2% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
efivarfs         24K   13K  6.7K  65% /sys/firmware/efi/efivars
/dev/sda1       1.1G  6.2M  1.1G   1% /boot/efi
tmpfs           370M  152K  370M   1% /run/user/1000
alish@alish:~$ 
```

20**. ping:**

Checks the connectivity between your machine and a remote host or IP address.

Command : ping <ipconfig>

```
alish@alish:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=33.6 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=39.0 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=115 time=32.8 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=115 time=33.5 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=115 time=50.8 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=115 time=1238 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=115 time=262 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=115 time=30.9 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=115 time=33.4 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=115 time=32.1 ms
```

21. **nslookup (Name Service Lookup)**

This command is used to query Domain Name System (DNS) details. It helps you find the IP address associated with a domain name or find the domain name associated with an IP address.

Command: nslook <name or ip>

```
alish@alish:~$ nslookup  instagram.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   instagram.com
Address: 57.144.48.34
Name:   instagram.com
Address: 2a03:2880:f311:22:face:b00c:0:4420

alish@alish:~$
```